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Dos & Ddos Attack

Man in The Middle Attack



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Outline

- Types of attacks
- What is Dos and Dos attack?
- Types of DoS Attack
- Symptoms of attack and preventative measures
- Man in the Middle Attack





Common Types of attacks

- Eavesdropping
- Data Modification
- Identity Spoofing (IP Address Spoofing)
- Password-Based Attacks
- **Denial-of-Service Attack (DoS)**
- **Man-in-the-Middle Attack**
- Compromised-Key Attack
- Application-Layer Attack
- Sniffer Attack

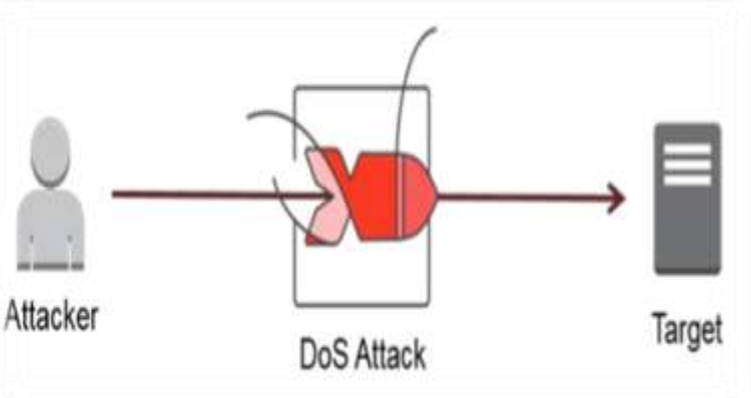
4 What is Dos and DDos attacks?



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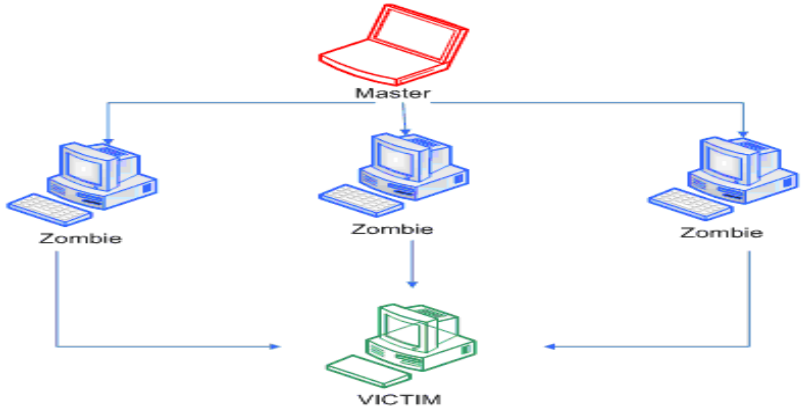
DoS

- In a denial-of-service (DoS) attack, an attacker attempts to prevent legitimate users from accessing information or services.



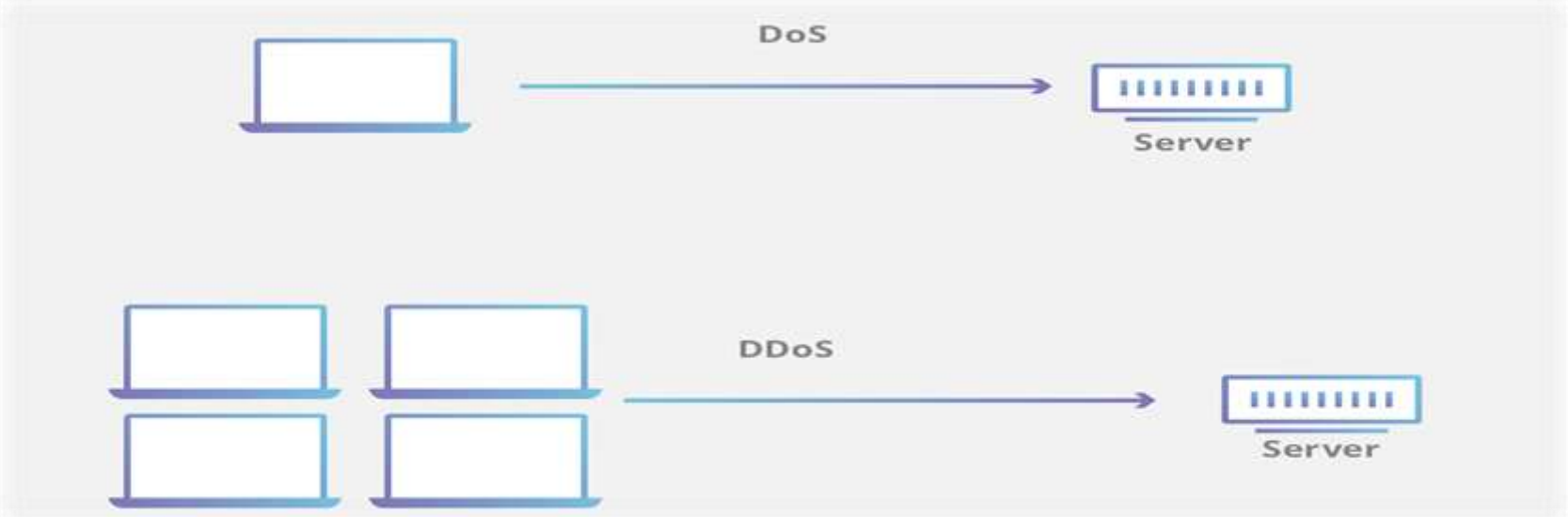
DDoS

- In a distributed denial-of-service (DDoS) attack, an attacker may use your computer to attack another computer.



6 Difference of DoS and DDoS Attacks

- DoS = when a single host attacks
- DDoS = when multiple hosts attack at the same time



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Types of DoS Attack

Some of the most commonly used DDoS attack types include:

- Penetration
- Eavesdropping
- Man-In-The-Middle
- Flooding



Penetration

- Attacker gets inside your machine
- Can take over machine and do whatever he wants
- Achieves entry via software flaw(s), stolen passwords or insider access



Eavesdropping

- Attacker gains access to same network
- Listens to traffic going in and out of your machine



Man-In-The-Middle

- Attacker listens to output and controls output
- Can substitute messages in both directions

Flooding

- Attacker sends an overwhelming number of messages at your machine; great congestion
- The congestion may occur in the path before your machine
- Messages from legitimate users are crowded out
- Usually called a Denial of Service (DoS) attack, because that’s the effect
- Usually involves a large number of machines, hence Distributed Denial of Service (DDoS) attack

Symptoms of attack and preventative measures

- Unusually slow network performance, unavailability of a particular website, inability to access any website, and dramatic increase in the amount of spam you receive.
- Symptoms could stem from hardware or software problems and mistaken for a DoS.
- Users can take steps to prevent becoming part of a botnet:
  - Install and use Anti-virus software
  - Setup a firewall to protect your system from unauthorized access
  - Use common sense security practices to avoid malware, trojans and viruses



Man in the Middle Attack

Public key cryptography is generally used in two ways:

1. by generating a shared secret at both ends of the communications link (key agreement)

   Diffie-Hellman key agreement scheme

2. by sending a secret to the other end of the communications link (key transport)

   RSA (Rivest, Shamir, Adleman) scheme

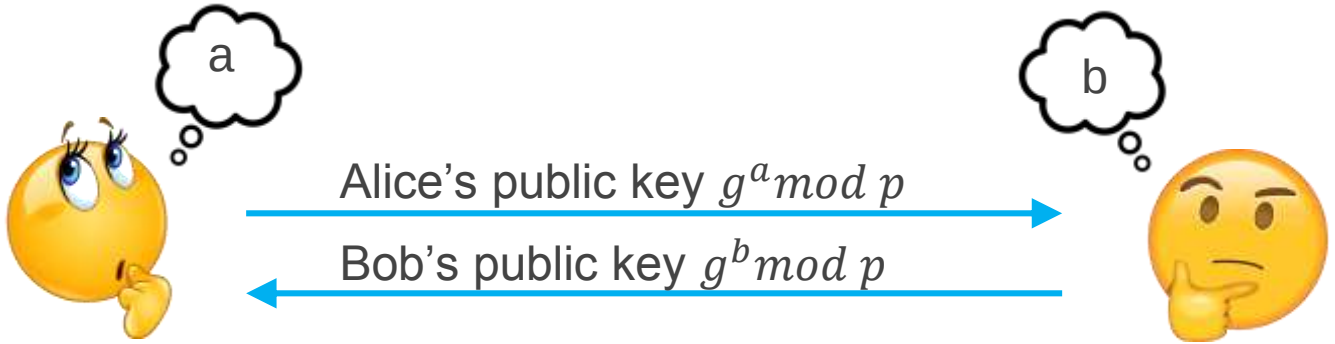
Alice and Bob knows p and g



Steps in the algorithm:

1. Alice and Bob agree on a prime number p and a base g.
2. Alice chooses a secret number a, and sends Bob $(g^a \mod p)$.
3. Bob chooses a secret number b, and sends Alice $(g^b \mod p)$.
4. Alice computes $((g^b \mod p)^a \mod p)$.
5. Bob computes $((g^a \mod p)^b \mod p)$.

Both Alice and Bob can use this number as their key. Notice that p and g need not be protected.



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Shared key $(g^b mod\ p)^a mod\ p$



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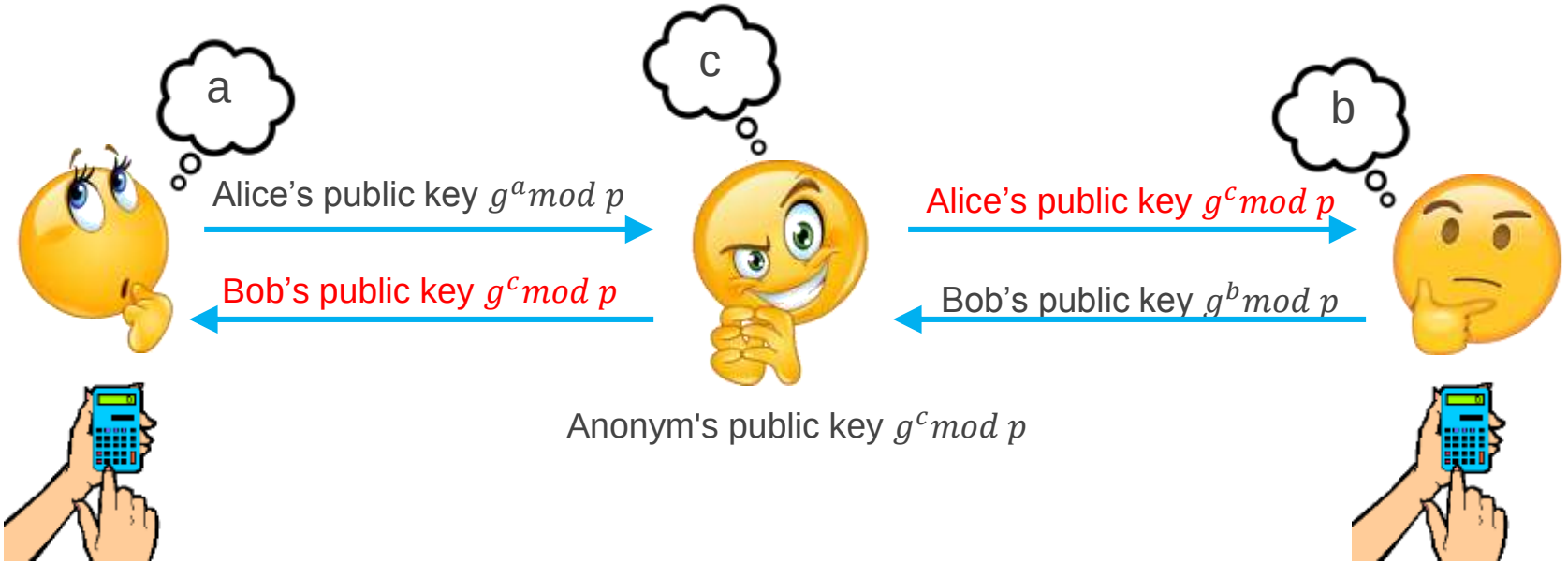


Key agreement schemes are vulnerable to man-in-the-middle attacks

What is Man-in-the-Middle Attack?

- In cryptography and computer security, a **man-in-the-middle attack** (**MITM**) also known as “**hijacking**” is an **attack** where the attacker secretly relays and possibly alters the communication between two parties who believe they are directly communicating with each other.

Man-in-the-Middle (MITM) Attack Concept





Man-in-the-middle attack types

Passive		*Active*
A passive attack is one in which the attacker is only able to monitor the communications channel. *Threatens confidentiality*		An active attack is one in which the attacker attempts to add, delete, or modify messages. Threatens both confidentiality and data integrity

…EXAMPLES…

Wi-Fi Eavesdropping

The hacker can create a fake Wi-Fi node disguised as a legitimate Wi-Fi access point to steal the personal information of everyone who connects.

Email Hijacking

The hacker gain access to important email accounts, and he will monitor the transactions. For example, they can wait for a scenario where the customer will be sending money and respond, spoofing the company's email address, with their own bank details instead of the company's. This way, the customer thinks they're sending their payment to the company, but they're really sending it right to the hacker.

…EXAMPLES…(CONT)

- Session Hijacking

Once you log into a website, a connection between your computer and the website is established. Hackers can hijack your session with the website through numerous means. One popular option they use is stealing your browser cookies. In case you don’t know, cookies store small pieces of information that makes web browsing convenient for you. It can be your online activity, login credentials, pre-fill forms, and in some cases, your location. If they got hold of your login cookies, they can easily log into your accounts and assume your identity.



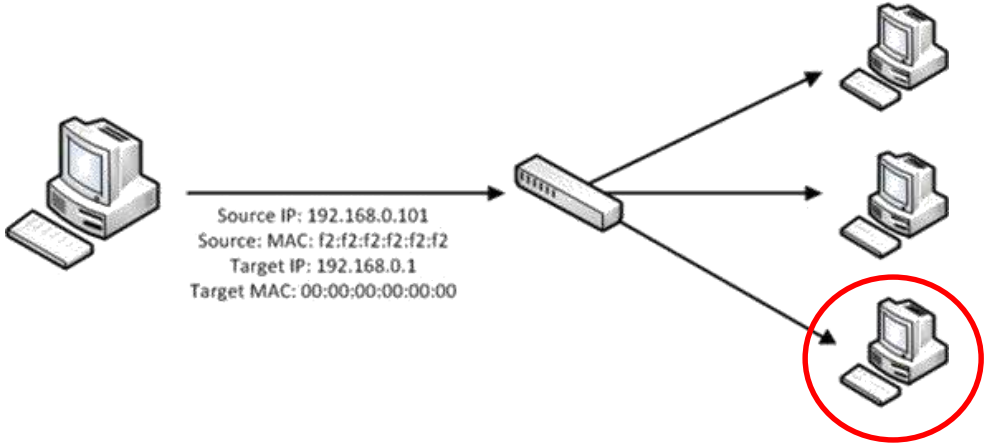
MITM TECHNIQUES

Man-in-the-middle attacks can be accomplished using a variety of methods

- **ARP poisoning-** ARP (Address Resolution Protocol) spoofing is also known as "ARP spoofing " or ARP Poison Routing. The attacker may use ARP spoofing to sniff data frames on LAN and to modify the packets. The attacker may corrupt the ARP caches of directly connected hosts and finally take over the IP address of the victim host. This requires that the attacker be on the same Ethernet segment as either the victim or the host with which it is communicating.

ARP cache

IP Address	MAC Address
192.168.0.1	00:00:00:00:00:00



who is 192.168.0.1?

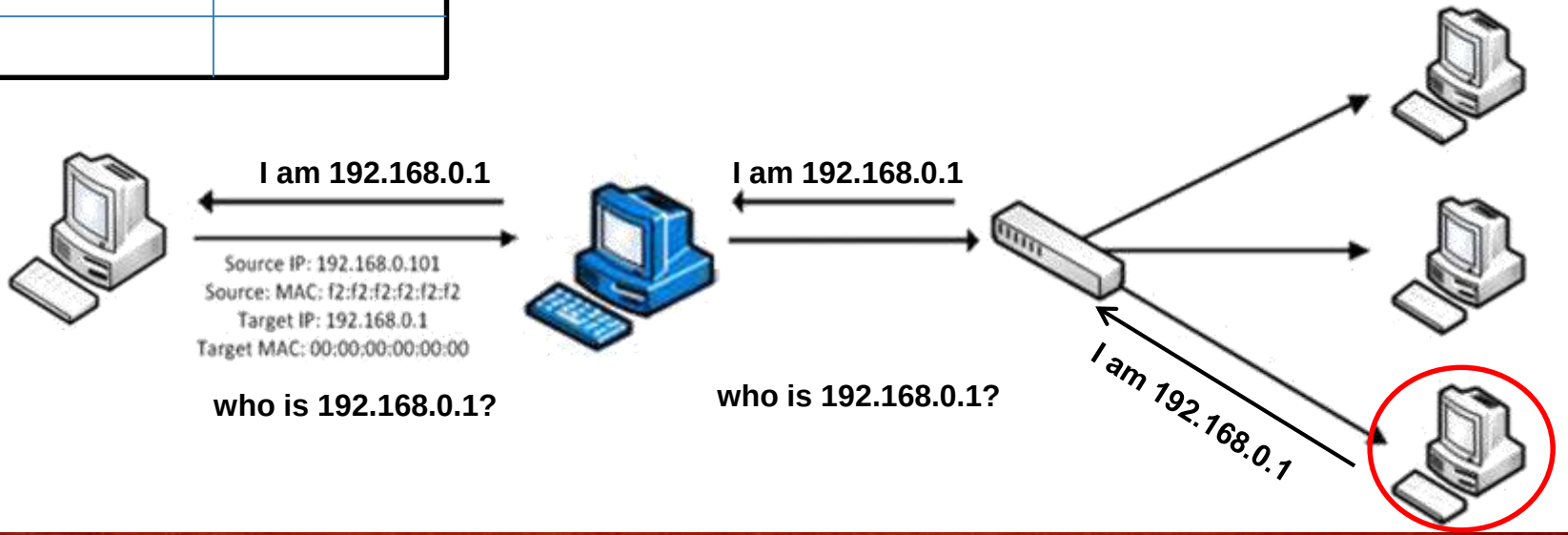
ARP request

I am 192.168.0.1

ARP reply

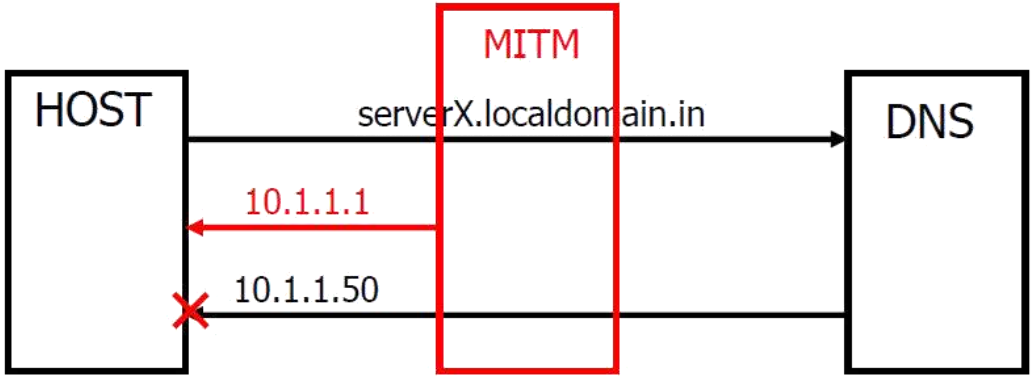
ARP cache

IP Address	MAC Address
~~192.168.0.1~~	~~00:00:00:00:00:00~~
192.168.0.1	f2:f2:f2:f2:f2:f2



Man-in-the-middle attacks can be accomplished using a variety of methods (CONT)

- **DNS spoofing-** The attacker starts by sniffing the ID of any DNS request, and then replies to the target requests before the real DNS server.





Defense and Detection

Encrypt, Encrypt, and Encrypt

- SSL/TLS Certificates will prevent you from connecting through the MitM.

If your website still uses the more vulnerable HTTP protocol, it's time to upgrade to the safer HTTPS protocol through SSL/TLS Certificates. A TLS Certificate will activate the HTTPS protocol, which is the safer version of HTTP. This allows an encrypted, secure connection between your server and your clients' computers, keeping all information from prying hackers.

- Tunnel into a trusted endpoint IPSEC(Internet Protocol security), SSH(Secure SHell) tunnels, VPN, Proxies
- Use the latest version of software (e.g. browsers)
- To stop ARP poisoning, use network switches that have MAC binding features. Switches with MAC binding store the first MAC address that appears on a port and do not allow the mapping to be changed without authentication.

- The second generation of the WPA security protocol (WPA2) is based on the final IEEE 802.11i amendment to the 802.11 standard. It uses AES encryption which makes it more secure compared to WEP and WPA protocols. The main vulnerability of WPA2 is the encryption done at WPA2 only is effective for who is kept outside the network.

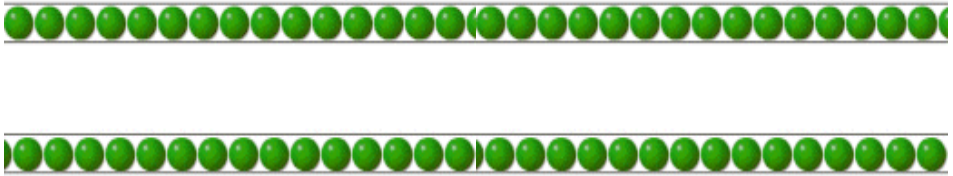


Demonstrating a Man in the Middle Attack using ARPspoof tool

Sender

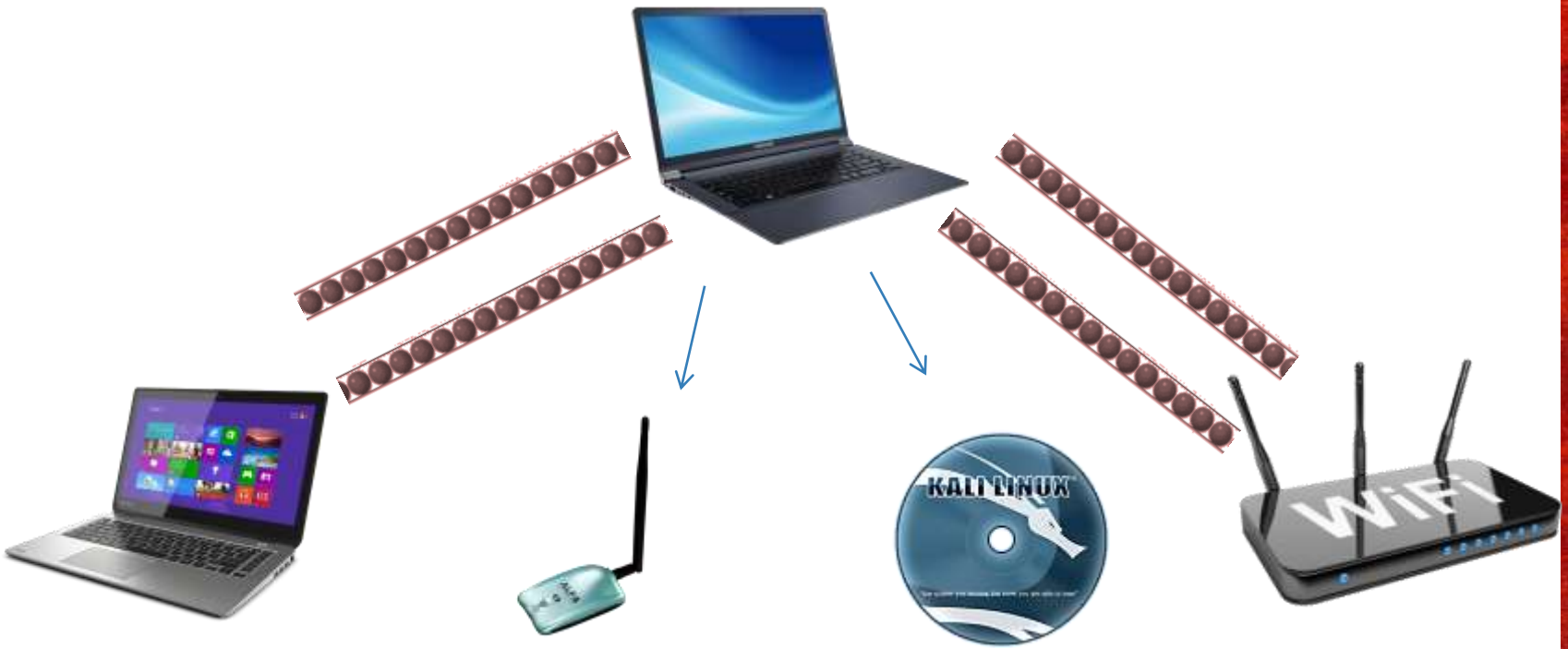
Receiver







What do we need?



Any Question

